C

LINEAR ALGEBRA • UNIT 3.01

Matrices, Matrix Algebra & Inverses

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 C	Q3 A	Q4 A	Q5 B
Q6 A	Q7 A	Q8 C	Q9 C	Q10 A
Q11 B	Q12 A	Q13 D	Q14 C	Q15 B
Q16 B	Q17 B	Q18 C	Q19 A	Q20 D
Q21 A	Q22 B	Q23 B	Q24 D	Q25 D

Matrices, Matrix Algebra & Inverses Practice Exercises

1 MCQ

Find $\|\mathbf{u}\|$ for $\mathbf{u} = (-9, -12)$.

A 21

B 14

C 15 ✓

D 225

WORKED SOLUTION & VERIFICATION

$\|\mathbf{u}\| = \sqrt{(-9)^2 + (-12)^2} = \sqrt{225} = 15.$

2 MCQ

A 3×3 symmetric matrix Q_k has eigenvalues $\lambda_1 = 12$, $\lambda_2 = 3$, and $\lambda_3 = 14$. Determine the trace $\text{tr}(Q_k)$.

A 26

B 32

C 29 ✓

D 41

WORKED SOLUTION & VERIFICATION

The trace equals $\lambda_1 + \lambda_2 + \lambda_3 = 29$.

3 MCQ

A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, find $T(\mathbf{v})$.

- **A** $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$ ✓
- **B** $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$
- **C** $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$
- **D** $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$

WORKED SOLUTION & VERIFICATION

To find $T(\mathbf{v})$, we multiply the matrix A by the vector $\mathbf{v}$. $T(\mathbf{v}) = A\mathbf{v} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Performing the matrix-vector multiplication: $T(\mathbf{v}) = \begin{pmatrix} (2 \times 1) + (1 \times 2) \\ (1 \times 1) + (3 \times 2) \end{pmatrix} = \begin{pmatrix} 2 + 2 \\ 1 + 6 \end{pmatrix} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$. Thus, the correct answer is $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$.

4 MCQ

Consider the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If a point $P = (x, y)$ is transformed by A to $P' = (x', y')$, such that $\begin{pmatrix} x' \\ y' \end{pmatrix} = A\begin{pmatrix} x \\ y \end{pmatrix}$, what is the image of the point $(3, 1)$ under this transformation?

- **A** (7, 6) ✓
- **B** (6, 7)
- **C** (8, 5)
- **D** (5, 8)

WORKED SOLUTION & VERIFICATION

The transformation is defined by the matrix multiplication $\begin{pmatrix} x' \\ y' \end{pmatrix} = A\begin{pmatrix} x \\ y \end{pmatrix}$. We are given the point $(3, 1)$, so $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. Performing the matrix multiplication:

$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} (2 \times 3) + (1 \times 1) \\ (1 \times 3) + (3 \times 1) \end{pmatrix} = \begin{pmatrix} 6 + 1 \\ 3 + 3 \end{pmatrix} = \begin{pmatrix} 7 \\ 6 \end{pmatrix}$

Therefore, the image of the point $(3, 1)$ is $(7, 6)$.

5 MCQ

Consider two vectors $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

- **A** 1
- **B** 0 ✓
- **C** -7
- **D** 25

WORKED SOLUTION & VERIFICATION

The dot product of two vectors $\mathbf{v}_1 = \begin{pmatrix} a \\ b \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} c \\ d \end{pmatrix}$ is given by $\mathbf{v}_1 \cdot \mathbf{v}_2 = ac + bd$.

In this case, $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$.

Therefore, $\mathbf{v}_1 \cdot \mathbf{v}_2 = (3)(-4) + (4)(3) = -12 + 12 = 0$.

A dot product of 0 indicates that the vectors are orthogonal.

6 MCQ

Consider the following system of linear equations represented by the augmented matrix:

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & -1 & 0 \end{array}\right)$$

. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x?

- **A** 1 ✓
- **B** 2
- **C** 3
- **D** 4

WORKED SOLUTION & VERIFICATION

We are given the augmented matrix:

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & -1 & 0 \end{array}\right)$$

To transform this matrix into row-echelon form, we can perform the following row operation: $R_2 \leftarrow R_2 - 2R_1$.

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2-2(1) & -1-2(2) & 0-2(5) \end{array}\right)$$

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -5 & -10 \end{array}\right)$$

Now, we can perform another row operation: $R_2 \leftarrow R_2/-5$.

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 1 & 2 \end{array}\right)$$

This matrix is now in row-echelon form. The corresponding system of equations is:

$x + 2y = 5$

$y = 2$

Substitute the value of y into the first equation:

$x + 2(2) = 5$

$x + 4 = 5$

$x = 1$

Therefore, the unique solution is $(1, 2)$, and the value of x is 1.

7 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$. Calculate the magnitude of $\mathbf{v}$, denoted as $\|\mathbf{v}\|$.

- **A** 13 ✓
- **B** 17
- **C** 7
- **D** 169

WORKED SOLUTION & VERIFICATION

The magnitude of a vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ in R^2 is given by the formula $\|\mathbf{v}\| = \sqrt{x^2 + y^2}$. For the given vector $\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$, we have $x = 5$ and $y = -12$. Therefore, $\|\mathbf{v}\| = \sqrt{(5)^2 + (-12)^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

8 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$. If this vector is plotted on a standard xy-plane, it represents a directed line segment from the origin to the point $(3, 4)$. What is the geometric interpretation of the length of this vector?

A The vector's direction in the xy-plane.

B The angle the vector makes with the positive x-axis.

C **The distance of the point (3,4) from the origin.** ✓

D The sum of the components of the vector.

WORKED SOLUTION & VERIFICATION

The length of a vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ is given by its magnitude, $\|\mathbf{v}\| = \sqrt{x^2 + y^2}$. Geometrically, this corresponds to the distance from the origin $(0, 0)$ to the point (x, y) on the Cartesian plane, as described by the Pythagorean theorem. For $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, the length is $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. This value represents the distance of the point $(3, 4)$ from the origin.

9 MCQ

A planar linear transformation B scales a by 21 and y by 14 with matrix $T = \begin{pmatrix} 21 & 0 \\ 0 & 14 \end{pmatrix}$. Determine the matrix trace $\operatorname{tr}(T)$.

A 38

B 19

C **35** ✓

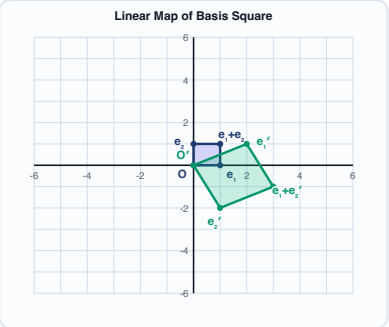
D 56

WORKED SOLUTION & VERIFICATION

The trace of the scaling matrix is the sum of its main diagonal entries: $\operatorname{tr}(T) = 21 + 14 = 35$.

10 MCQ

Consider a linear transformation represented by the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}$. A point P is mapped to P' by this transformation. If the coordinates of P are $\begin{pmatrix} x \\ y \end{pmatrix}$ and the coordinates of P' are $\begin{pmatrix} x' \\ y' \end{pmatrix}$, then $\begin{pmatrix} x' \\ y' \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix}$. Suppose a curve is defined by the equation $x'^2 + y'^2 = 5$. What is the equation of the original curve $x^2 + y^2 = k$ in terms of x and y that transforms into this circle? Find the value of k.



A 1 ✓

B 2

C 3

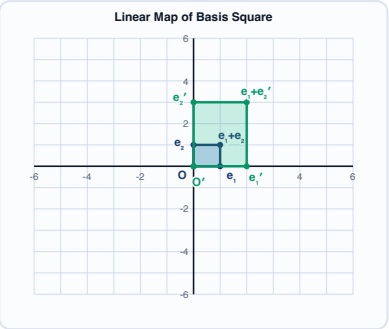
D 4

WORKED SOLUTION & VERIFICATION

The transformation is given by $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$. We can express $\begin{pmatrix} x \\ y \end{pmatrix}$ in terms of $\begin{pmatrix} x' \\ y' \end{pmatrix}$ by finding the inverse of matrix A. The determinant of A is $\det(A) = (2)(-2) - (1)(1) = -4 - 1 = -5$. The inverse matrix is $A^{-1} = \frac{1}{-5}\begin{pmatrix} -2 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 2/5 & 1/5 \\ 1/5 & -2/5 \end{pmatrix}$. Therefore, $\begin{pmatrix} x \\ y \end{pmatrix} = A^{-1}\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2/5 & 1/5 \\ 1/5 & -2/5 \end{pmatrix}\begin{pmatrix} x' \\ y' \end{pmatrix}$. This gives us $x = \frac{2}{5}x' + \frac{1}{5}y'$ and $y = \frac{1}{5}x' - \frac{2}{5}y'$. We are given the transformed equation $x'^2 + y'^2 = 5$. We need to substitute x' and y' in terms of x and y into this equation. From the transformation equations, we can write $x' = 2x + y$ and $y' = x - 2y$. Substituting these into $x'^2 + y'^2 = 5$: $(2x + y)^2 + (x - 2y)^2 = 5$. Expanding this: $(4x^2 + 4xy + y^2) + (x^2 - 4xy + 4y^2) = 5$. Simplifying, we get $5x^2 + 5y^2 = 5$. Dividing by 5, we have $x^2 + y^2 = 1$. Thus, the original curve is $x^2 + y^2 = k$ where $k = 1$.

11 MCQ

Consider the matrix $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. If $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, what is $A\mathbf{v}$?



A $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

B $\begin{pmatrix} 2 \\ 6 \end{pmatrix}$ ✓

C $\begin{pmatrix} 1 \\ 6 \end{pmatrix}$

D $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$

WORKED SOLUTION & VERIFICATION

To find $A\mathbf{v}$, we perform matrix multiplication.

$$A\mathbf{v} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} (2)(1)+(0)(2) \\ (0)(1)+(3)(2) \end{pmatrix} = \begin{pmatrix} 2+0 \\ 0+6 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}.$$

12 MCQ

Consider the matrix $A = \begin{pmatrix} k & 2 \\ 1 & -3 \end{pmatrix}$. If the matrix A is invertible and its inverse A^{-1} has a determinant of $-\frac{1}{11}$, find the value of k.

A 3 ✓

B 4

C 5

D 6

WORKED SOLUTION & VERIFICATION

The determinant of $A = \begin{pmatrix} k & 2 \\ 1 & -3 \end{pmatrix}$ is:

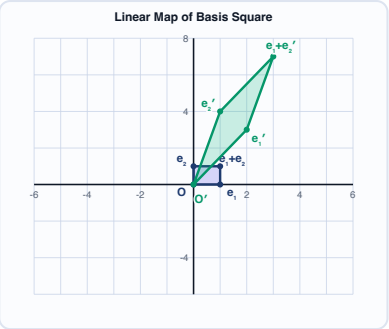
$$\det(A) = -3k - 2.$$

Since $\det(A^{-1}) = \frac{1}{\det(A)} = -\frac{1}{11}$, we have $\det(A) = -11$:

$$-3k - 2 = -11 \implies -3k = -9 \implies k = 3.$$

13 MCQ

Consider the matrix $A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$. What is the matrix A^T?



A $\begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix}$

B $\begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$

C $\begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}$

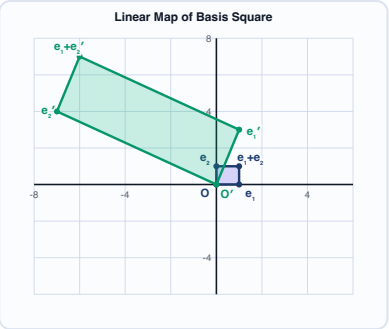
D $\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$ ✓

WORKED SOLUTION & VERIFICATION

The transpose of a matrix A, denoted by A^T, is obtained by interchanging its rows and columns. For a matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, its transpose is $A^T = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$. Given the matrix $A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$, interchanging the rows and columns yields $A^T = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$.

14 MCQ

Let A be a 2×2 matrix given by $A = \begin{pmatrix} 2 & k \\ 3 & -1 \end{pmatrix}$. If $A^2 = \begin{pmatrix} 1 & -7 \\ 3 & 4 \end{pmatrix}$, find the value of k.



A 2

B 4

C 3 ✓

D 5

WORKED SOLUTION & VERIFICATION

We are given the matrix $A = \begin{pmatrix} 1 & 2 \\ k & 3 \end{pmatrix}$. The transpose of A, denoted by A^T, is obtained by interchanging its rows and columns. Thus,

$$A^T = \begin{pmatrix} 1 & k \\ 2 & 3 \end{pmatrix}$$

We are given the equation $A + A^T = \begin{pmatrix} 2 & 5 \\ 5 & 6 \end{pmatrix}$. Let's compute the sum $A + A^T$:

$$A + A^T = \begin{pmatrix} 1 & 2 \\ k & 3 \end{pmatrix} + \begin{pmatrix} 1 & k \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 1+1 & 2+k \\ k+2 & 3+3 \end{pmatrix} = \begin{pmatrix} 2 & 2+k \\ k+2 & 6 \end{pmatrix}$$

Now, we equate this resulting matrix to the given matrix $A + A^T = \begin{pmatrix} 2 & 5 \\ 5 & 6 \end{pmatrix}$:

$$\begin{pmatrix} 2 & 2+k \\ k+2 & 6 \end{pmatrix} = \begin{pmatrix} 2 & 5 \\ 5 & 6 \end{pmatrix}$$

By comparing the corresponding entries of the matrices, we can find the value of k. From the entry in the first row and second column (or the second row and first column), we have:

$$2 + k = 5$$

Solving for k:

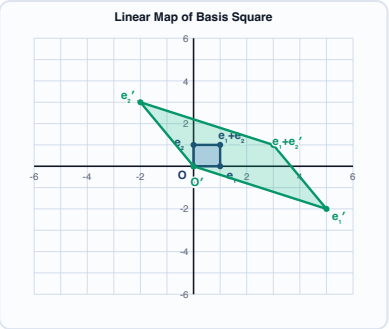
$$k = 5 - 2$$

$$k = 3$$

The other entries are consistent: the (1,1) entry is $2 = 2$, and the (2,2) entry is $6 = 6$. Therefore, the value of k is 3.

15 MCQ

For $A = \begin{bmatrix} 5 & -2 \\ -2 & 3 \end{bmatrix}$, find entry $(-3A)_{12}$.



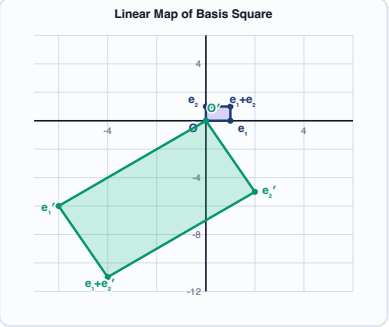
A -2	B 6 ✓
C -6	D 7

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(-3A)_{12} = -3(-2) = 6$. Therefore, the verified correct option is 6.

16 MCQ

For $A = \begin{bmatrix} -6 & 2 \\ -6 & -5 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -2 \\ 4 & 5 \end{bmatrix}$, find entry $(A + B)_{12}$.



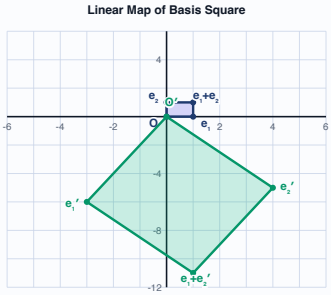
A 2	B 0 ✓
C -2	D 1

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(A + B)_{12} = (2) + (-2) = 0$. Therefore, the verified correct option is 0.

17 MCQ

For $A = \begin{bmatrix} -3 & 4 \\ -6 & -5 \end{bmatrix}$, $B = \begin{bmatrix} 6 & -1 \\ 2 & -5 \end{bmatrix}$, find entry $(A+B)_{22}$.



A -5

B **-10** ✓

C -9

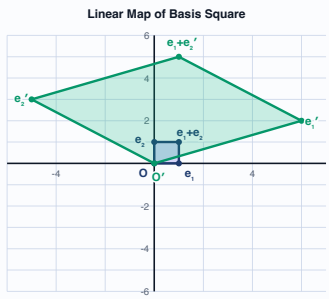
D -12

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(A+B)_{22} = (-5)+(-5) = -10$. Therefore, the verified correct option is -10.

18 MCQ

For $A = \begin{bmatrix} 6 & -5 \\ 2 & 3 \end{bmatrix}$, find entry $(-3A)_{22}$.



A 3

B 9

C **-9** ✓

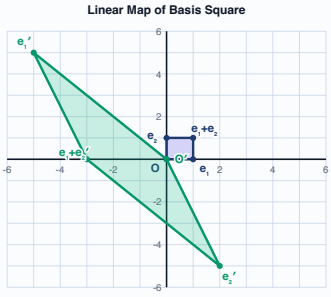
D -8

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(-3A)_{22} = -3(3) = -9$. Therefore, the verified correct option is -9.

19 MCQ

For $A = \begin{bmatrix} -5 & 2 \\ 5 & -5 \end{bmatrix}$, find entry $(2A)_{22}$.



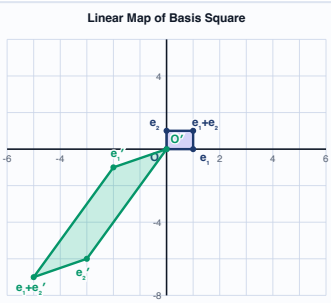
A -10 ✓	**B** -5
C 10	**D** -9

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(2A)_{22} = 2(-5) = -10$. Therefore, the verified correct option is -10.

20 MCQ

For $A = \begin{bmatrix} -2 & -3 \\ -1 & -6 \end{bmatrix}$, find entry $(-3A)_{22}$.



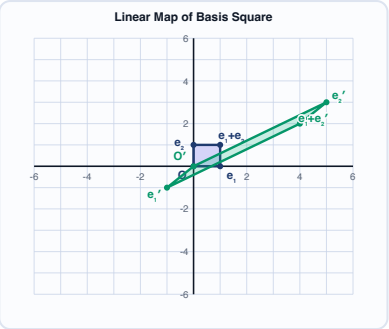
A 19	**B** -6
C -18	**D** 18 ✓

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(-3A)_{22} = -3(-6) = 18$. Therefore, the verified correct option is 18.

21 MCQ

For $A = \begin{bmatrix} -1 & 5 \\ -1 & 3 \end{bmatrix}$, find entry $(-4A)_{21}$.



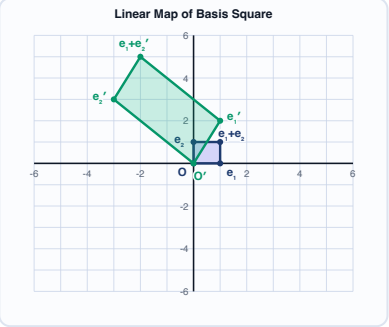
- **A** 4 ✓
- **B** 5
- **C** -1
- **D** -4

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(-4A)_{21} = -4(-1) = 4$. Therefore, the verified correct option is 4.

22 MCQ

For $A = \begin{bmatrix} 1 & -3 \\ 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 2 \\ 6 & -1 \end{bmatrix}$, find entry $(A + B)_{21}$.



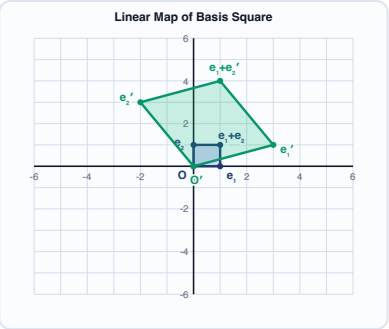
- **A** 2
- **B** 8 ✓
- **C** 6
- **D** 9

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(A + B)_{21} = (2) + (6) = 8$. Therefore, the verified correct option is 8.

23 MCQ

For $A = \begin{bmatrix} 3 & -2 \\ 1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} -3 & -6 \\ -6 & 4 \end{bmatrix}$, find entry $(A+B)_{11}$.



A 3

B 0 ✓

C -3

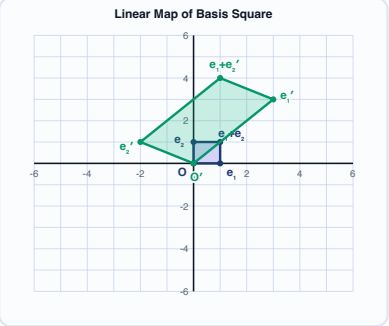
D 1

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(A+B)_{11} = (3)+(-3) = 0$. Therefore, the verified correct option is 0.

24 MCQ

For $A = \begin{bmatrix} 3 & -2 \\ 3 & 1 \end{bmatrix}$, find entry $(-3A)_{12}$.



A -2

B -6

C 7

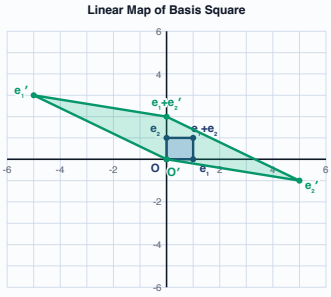
D 6 ✓

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(-3A)_{12} = -3(-2) = 6$. Therefore, the verified correct option is 6.

25 MCQ

For $A = \begin{bmatrix} -5 & 5 \\ 3 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 6 & 1 \\ -3 & -6 \end{bmatrix}$, find entry $(A+B)_{22}$.



A -6

B -9

C -1

D **-7** ✓

WORKED SOLUTION & VERIFICATION

Applying Matrices & Matrix Algebra principles, $(A+B)_{22} = (-1)+(-6) = -7$. Therefore, the verified correct option is -7.